BRIAN OTINA MAKAU

Data Engineer | Analyst

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Professional Summary

I have been working as a Data Analyst for the past several years, I am proficient with Python and SQL to automate data quality tasks, data transformation and enhance business workflows. Possesses a Bachelor of Science in Applied Statistics with Programming and has had some exposure to working with machine learning and statistical modelling. Passionate about leveraging software engineering and automation techniques for AI powered solutions, LLM integrations, and intelligent business workflows. Also, I have been developing, deploying, and maintaining web applications, back-end APIs, and automated data workflows. Familiar with Python, Django REST Framework, Node.js, TypeScript, SQL, and Git, and has experience building REST APIs, modeling databases, maintaining applications, and automating data pipelines. Built a full stack web application and back-end projects with authentication, authorization, database operations, and modular architecture of services.

Professional Experience

Business Data Analyst, *MultiChoice* 05/2025 – 09/2025
Ngong road, KENYA

- Automated reporting and data-processing workflows, contributing to a 10% reduction in average turnaround time.
- Cleaned, transformed, and organized installer data for migration to a new system, reducing the time required for the process by 20%.
- Analyzed operational data and supported data-driven improvements that reduced refund-processing time by 5%.
- Collaborated with business stakeholders to support reporting improvements and operational decision-making.
- Applied analytical thinking and data preparation techniques to improve the usability of business information.

Big Data Engineer Intern, *Safra Data* 04/2024 – 08/2024
Westland, KENYA

- Automated data profiling and quality checks using Python, improving dataset accuracy by 15%.
- Optimized SQL queries and Apache Airflow workflows to improve data-processing performance.
- Applied partitioning, bucketing, salting, and caching techniques to improve data-processing efficiency.
- Worked with data transformation and workflow orchestration to support reliable data delivery.
- Investigated data-quality issues and applied analytical problem-solving to improve processing workflows.

Software Developer, *Paza Social* 01/2026 – 07/2026
Nairobi, KENYA

Full-Stack Development, Deployment and Maintenance | Less than 6 months Website: https://paza.social Technologies: Next.js, TypeScript, Node.js, Prisma ORM, Amazon S3, Git.

- Developed and maintained frontend and backend components of a live web application using Next.js, TypeScript, and Node.js.
- Implemented and maintained modular backend functionality using API controllers, routes, and service modules to support application features and business logic.
- Worked on API functionality for discovery, messaging, lead management, outreach search, pipeline health, and administrative operations.

- Managed application data models using Prisma ORM, database schemas, and migrations to support structured data access and database changes.
- Worked with an Amazon S3 upload service to support file-upload functionality and cloud service integration.
- Organized backend code into controllers, routes, services, shared modules, and type definitions to improve maintainability and separation of concerns.
- Deployed and maintained the live application, contributing to troubleshooting, ongoing improvements, and system reliability.

Personal Projects

Cryptocurrency Data Pipeline
Technologies: Python, Apache Kafka, Apache Airflow, Docker, Amazon S3, Snowflake, Snowpipe, Schema Registry, SQL, Power BI, Hadoop
- Designed a data ingestion architecture using Apache Kafka for streaming and Apache Airflow for workflow orchestration.
- Implemented a Medallion Architecture with Bronze, Silver, and Gold layers to organize raw, cleansed, and analytics-ready data.
- Used Amazon S3 for raw data storage and integrated Snowflake and Snowpipe for automated data loading and analytical querying.
- Containerized pipeline components with Docker to support consistent development and deployment environments.
- Incorporated schema management into the streaming architecture to support data consistency and structured processing.
- Prepared data for downstream SQL analytics and Power BI reporting, supporting analysis of cryptocurrency data.
- Designed the pipeline to separate ingestion, storage, transformation, and visualization responsibilities for maintainability and scalability.

Data Warehousing and Analytics
Technologies: SQL Server, SQL, Python, Power BI
- Designed a data warehousing workflow to consolidate structured data for reporting and analytical use.
- Used SQL Server to store and query data and support the preparation of datasets for reporting.
- Applied ETL processes to extract, clean, transform, and prepare data for consistent analytical consumption.
- Developed data models and organized datasets to support business intelligence and reporting requirements.
- Built Power BI dashboards featuring KPI cards, charts, and tabular views to communicate trends and performance indicators.
- Developed visualizations for measures including total runtime and ROI, vote counts by genre, profitability and budget, and budget by release period.
- Connected data preparation and modeling with dashboard development to support interactive exploration and data-driven analysis.

Airflow Orchestration, *Automated Data Pipeline*
Technologies: Python, Apache Airflow, Docker, Amazon S3, Snowflake, SQL, Reddit API, ExchangeRates API, Gmail notifications
- Developed an orchestrated data pipeline using Apache Airflow to coordinate ingestion, transformation, storage, and downstream analytics.
- Created Python-based workflow tasks and scheduled DAGs to automate data processing.
- Integrated external data sources, including the Reddit API and ExchangeRates API, to demonstrate multi-source data ingestion.
- Used Amazon S3 for intermediate or processed data storage and Snowflake for structured analytical data.
- Incorporated SQL-based data preparation to support downstream analysis and reporting.
- Used Docker to containerize pipeline components and support a consistent execution environment.
- Designed inter-task dependencies to coordinate workflow execution and manage processing order.
- Integrated email notifications to provide status updates when pipeline runs completed or encountered issues.
- Connected processed data to Power BI to support reporting and visualization.

Education

Bachelor of Science in Applied Statistics with Programming, 09/2020 – 04/2026
Murang'a University of Technology KENYA
Relevant Coursework: Python and R Programming, Database Systems, Statistical Inference, Applied Regression Analysis, Time Series Analysis, Machine Learning Fundamentals, Data Visualization and Reporting.

Technical Skills

Programming Languages
Python, TypeScript, JavaScript, SQL, R, Shell scripting.

Backend Development and APIs
Django, Django REST Framework, Node.js, RESTful APIs, API controllers and routes, service-layer architecture, CRUD operations, token-based authentication, authorization, permissions, filtering, search, and request throttling.

Automation and Data Pipelines
Python scripting, SQL automation, ETL pipelines, Apache Airflow, workflow orchestration, data validation, data cleaning, and data transformation.

Development and Deployment Tools
Git, GitHub, Docker, Linux, Amazon S3, cloud service integration, environment configuration, and application deployment and maintenance.

Software Engineering
Object-oriented programming fundamentals, modular design, reusable components, debugging, application maintenance, code organization, error handling, and technical problem-solving.

Web Development
Next.js, React, Vite, Tailwind CSS, frontend-backend integration, and full-stack application development.

Databases and Data Access
PostgreSQL, SQL Server, SQLite, Prisma ORM, relational data modeling, database schemas, migrations, and SQL queries.

Documentation and Collaboration
Technical documentation, analytical reporting, stakeholder communication, troubleshooting, teamwork, and continuous learning.

Certifications

Oracle Cloud Infrastructure 2025 Certified Foundations Associate

Oracle Data Platform 2025 Certified Foundations Associate

Oracle Autonomous Database Cloud 2025 Professional

Programming in Python
Computer Pride

Cognitive.ai powered by IBM Data Analysis with Python

AWS Certified Data Engineer – Associate
In Progress

Additional Strengths

- Analytical thinking and structured problem-solving
- API development and integration
- Technical documentation and collaborative development
- Automation of repetitive data and reporting workflows
- Application troubleshooting and maintenance
- Interest in LLM applications, AI tools, retrieval pipelines, and agentic automation